Introduction:

The opioid crisis began in the 1950s when doctors were overprescribing opioid painkillers. As doctors stopped prescribing opioids in response to rising addiction and overdoses, synthetic opioids hit the market. Since then, we've seen a sharp increase in opioid-related overdoses as drugs are being formulated to be increasingly potent, most notably with the rise in fentanyl overdoses. Opioids are respiratory depressants. They can slow down your breathing to a near halt, depriving your brain of oxygen. The opioid crisis has plagued America for nearly 70 years, and only in the past few years have we seen a slight decline. While that is progress, it is not enough. I want to find where the system is failing these people and how we can be more proactive in helping high-risk individuals within our county. To do so, I will analyze nationwide data and bring it into the context of Montgomery County. Concentrated effort tailored to cities can create a much more meaningful impact than a one size fits all blanket approach.

I hope to accomplish this using the State Unintentional Drug Overdose Reporting System (SUDORS). SUDORS has 274 columns, each detailing a rate or percentage of all unintentional overdoses in a state and year that falls into a certain category or circumstance. The data spans from 2020 to 2024 and includes all 50 states. I will also use the National Survey on Drug Use and Health (NSDUH). NSDUH surveys approximately 70,000 random people across the United States regardless of drug use status every year. This data includes surveys from 2021 to 2024. I will support these core sources with the CDC's Naloxone dispensing rate. This looks at distribution in the United States of retail pharmacies dispensed naloxone prescriptions per 100 persons per year from 2019 to 2024, as well as DataMontgomery's Opioid Crime data detailing exact locations of arrests collected from police reports from 2017 to present.

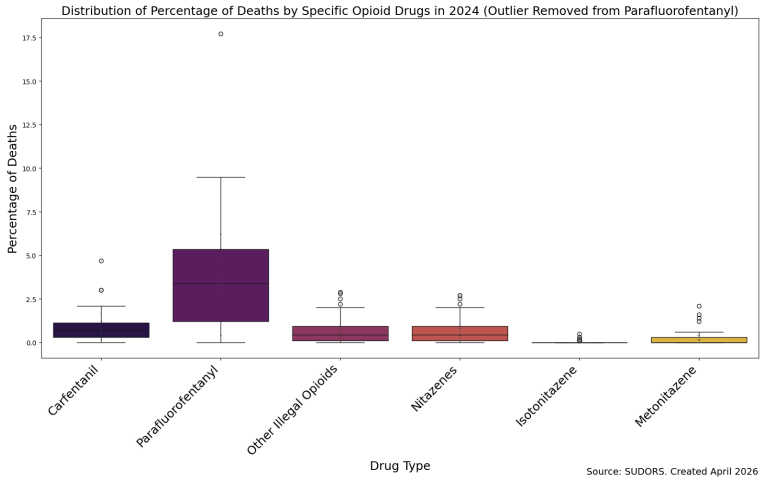
I will use Python for all of my analyses. I used the Pearson correlation test with the SUDORS dataset to test for significance of drug type in opioid overdoses. Then I conducted a Chi-square Test of Independence with the NSDUH dataset to look for significance of certain markers of an at-risk person in contributing to whether or not someone misused opioids in the past year. I then conducted an OLS model to find significant circumstances surrounding an overdose using SUDORS data. Next, I conducted a Pearson coefficient test to look for a significant relationship between naloxone distribution rates and overdoses. Finally, I mapped out the frequency of opioid possession and sale arrests in Montgomery County to look for points of interest.

Cleaning

The NSDUH dataset is so large it would either slow everything down or not load at all, so the first thing I needed to do was filter for only the necessary columns and rows. I went through the data dictionary and selected only the necessary rows that are often associated with drug use or indicate some type of disparity. I also filtered for only people who have misused opioids in the past year. I labeled this new data filtered_nsduh. There are numeric codes for skips or NAs that I cleaned out when using the dataset instead of in the cleaning notebook because I wanted to leave the option of looking at how many people skipped the question or the person taking the survey answered "don't know." Ultimately, I didn't end up incorporating that, so the codes for missing values just got converted to NAs and were dropped. I filtered through the SUDORS dataset for 18 circumstances surrounding the overdose. Finally, I added the overdose death rate for each state from SUDORS to more easily compare dispensing rates to death rates.

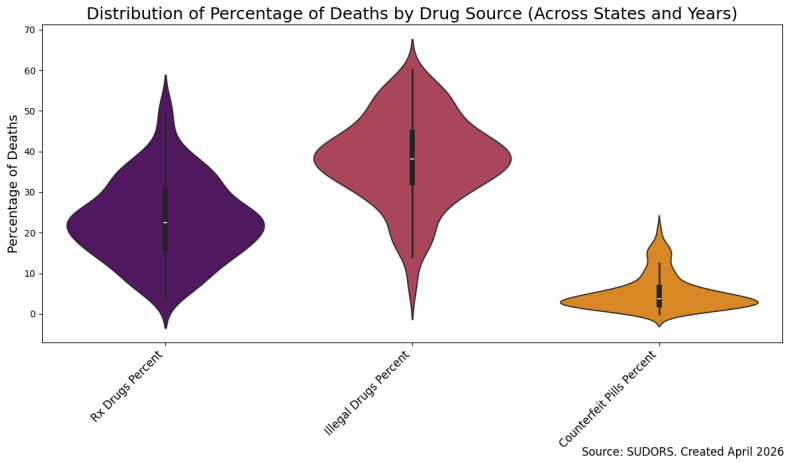
EDA

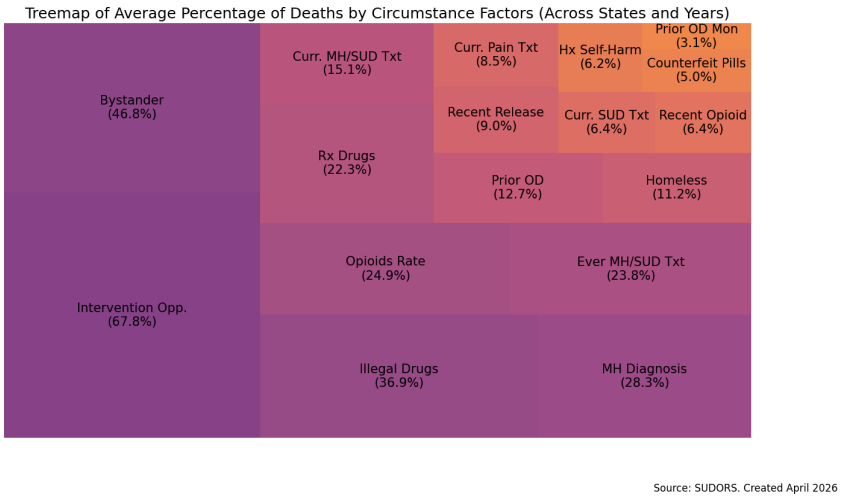
First, I wanted to look at the distribution of types of opioids and sources. Then, I wanted to determine the most frequent circumstances related to overdoses.



Para-fluorofentanyl and carfentanil are both analogs of fentanyl. Many people do not intentionally take these drugs. They often lace drugs like heroin to make them more addictive and add more mass for less. I am not surprised these are the most frequent opioids resulting in overdose because people likely don't know what's happening to them or how to react.

Nitazene is a potent painkiller that never made it to market. Isotonitazene and metonitazene are both analogs of nitazene. Illegal drugs appear to contribute to the most overdoses, next to Rx, or prescription drugs. This seems surprising considering all the specified opioids in SUDORS are illegal opioids. Counterfeit appears to be the least common opioid source. Nitazenes and their analogs can be taken in the form of counterfeit pills made to mimic real prescriptions.





Here I am looking at what percentage of unintentional overdoses satisfy these circumstances. Nearly half of overdoses have a bystander present, and over half of unintentional overdoses had an opportunity for help. This indicates a failing in the system. 67.8% of people who died of an overdose had an opportunity to prevent it, but that opportunity proved to be unsuccessful.

These descriptive analyses give us a better idea of some patterns existing in who is overdosing and from what. With this context, we will be able to gain more insight into the significance of some of these factors in the rate of overdose.

Analysis

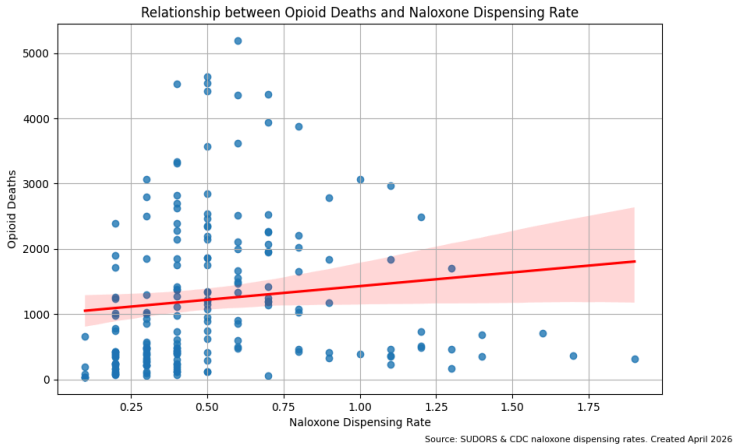
For my first test, I will be answering the question, "What type of opioids are people dying from the most and how are they obtained?" with a Pearson Correlation test. My null hypothesis is that there is no significant linear correlation between unintentional overdose and a certain drug, and my alternate hypothesis is that there is a significant linear correlation between unintentional overdose and a certain drug. They all came out to have a p-value between .9 and 1. This shows an extremely insignificant correlation. That means I cannot reject the null hypothesis. As to how they are obtained, there is no significant drug to focus on, but the violin plot gives us an idea of distribution and frequency, with illegal opioids being the most frequent, then Rx opioids, and then the least frequent source being counterfeit pills. This was shocking to me considering how many more para-fluorofentanyl deaths there were in my boxplot. Frequency does not equal significance, so I can conclude that if we were to focus on eradicating an opioid from the market, there likely would not be much of an impact on overdose rate.

My next test answered the question, "What are the markers of an at-risk person to fall into opioid use disorder?" with a Chi-square Test of Independence. This tests for significant association between two categorical variables. My variables were originally stored as 1 for yes and 2 for no, so first I converted those to categorical values. Then I tested 35 columns, or markers, against the variable OPINMYR, which indicates whether the respondent has misused opioids in the past year. My null hypothesis is that there is no significant relationship between a specific marker and misusing opioids, and my alternate hypothesis is that there is a significant relationship between a specific marker and misusing opioids. Eight variables satisfied my alternate hypothesis: arrested for drug possession or sale (p-value: 0.00), health score (p-value: 0.00), family receives any government assistance (p-value: 0.00), family receives food stamps (p-value: 0.00), arrested for a DUI (p-value: 0.02), family receives public assistance (p-value: 0.02), wouldn't work for a company that tests pre-hiring (p-value: 0.02), and family receives welfare or childcare (p-value: 0.04). There is a big trend between people who misuse opioids and receive any form of government assistance. There is also a significant relationship between opioid misuse and having been arrested for drug possession or sale or a DUI. Targeting and providing some type of treatment or prevention resources to people in custody regarding substances could make an impact on addiction and overdose rate, provided the method is effective.

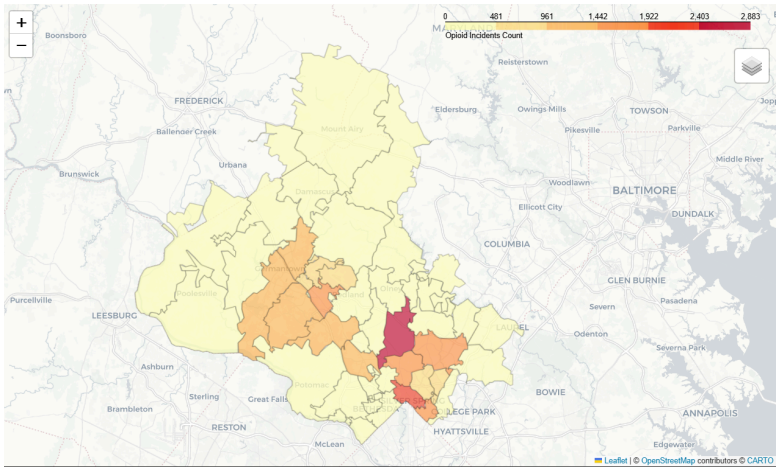
I then conducted an OLS model to answer the question, "What are common circumstances surrounding overdose deaths?" I compared 15 circumstances against the rate of unintentional opioid overdoses. I found 7 significant variables: ever previously overdosed (p-value: 0.00), previously overdosed within a month (p-value: 0.00), currently in SUD treatment (p-value: 0.00), evidence of illegal drug use (p-value: 0.01), ever been in mental health or SUD treatment (p-value: 0.02), recently released from a treatment facility (p-value: 0.02), and had an opportunity for prevention (p-value: 0.02). This continues to reinforce the conclusion that there is a significant failing in the system. The three lowest p-values are for variables that indicate they were likely in the care of a facility or hospital. Despite being in a situation to get help for their substance use disorder, these people still went on to die of an overdose. All the indications for being in treatment, whether that is at any point, recently, or currently, have a significant relationship with people who died of an opioid overdose. Of course, this could reflect more on

the severity and duration of the patients' addiction. We must still work to improve the system so people who receive treatment are actually able to stop using opioids.

When discussing solutions to the opioid crisis, naloxone, better known as Narcan, is always suggested as a life saving intervention that blocks and reverses opioids action on the brain. I wanted to see if naloxone distribution actually has a significant impact on lethal overdose rates. I conducted a Pearson Correlation test to look for significant correlation. My null hypothesis is that there is no significant correlation between lethal opioid overdose rate and naloxone distribution, and the alternate hypothesis is that there is a significant correlation between lethal opioid overdose rate and naloxone distribution. The p-value came out to be 0.1, meaning I failed to reject the null hypothesis. There is no significant correlation. I plotted the regression line, and as you can see, the points are quite random. There is a slight positive relationship indicating naloxone is being distributed more in response to overdoses, but distribution is not bringing the rate of overdoses down. This may not be fully indicative of naloxone failing, but rather that the line is relatively horizontal because states are generally meeting their needs at the right amount. A negative relationship sounds ideal when looking at high distribution rates being met with low overdose rates. However, that would require states with high overdoses having some of the lowest distribution rates. While that would allow us to better show a relationship, it is a good sign there are not a significant amount of states seeming to fail to combat opioid overdoses. SUDORS indicated that 22% of overdose deaths were administered naloxone prior to dying. Naloxone reports to be 75 to 100% effective, so this number is within normal range. That 22% could be attributed to user error, administering the treatment too late, the overdose being nonopioid related, or the person not immediately going to the hospital after.



Finally, I wanted to put these findings into the context of Montgomery County. I used GIS in Python to create an interactive map that will tell you how many opioid-related arrests have been made in a given zip code and city since 2017. Both red zones and the surrounding orange zones to the east side belong to Silver Spring. For some reason, one of the sources I included does not recognize Wheaton, but it is the orange section between the two red zones. The next hotspots occur around Rockville,



Gaithersburg, and Germantown. According to Data Commons out of 50 cities in Montgomery County, Silver Spring is ranked 47th in median household income, Wheaton is 43rd, Gaithersburg is 41st, Germantown is 37th, and Rockville is 32nd. This seems to support the correlation we found between opioid misuse and government assistance. This suggests that these areas should receive more funding and resources to provide care and help to those they arrest for opioid possession/sale and DUIs because they are especially high risk for a lethal overdose.

Suggested Applications

To determine next steps, we first have to look at what is already in place. I have already discussed naloxone distribution. While it is important and saves a lot of lives, it does not have much correlation to opioid overdoses, so efforts would be better focused elsewhere. More education about the use of naloxone could help decrease the 22% death rate associated with naloxone treatment. Lots of educational material and PSAs are already out there. Montgomery County has numerous resources available to help you learn how to act when you are around someone who is overdosing and how to prevent an overdose yourself. The United States has focused efforts on combating opioids at the source and cutting access. The United States, along with Maryland, is trying to expand access to treatment by creating more facilities. Based on my findings, this could be ineffective without the proper care and initiative to reform treatment facilities. As we saw, a significant number of people who receive treatment at facilities still succumbed to opioids, losing their lives.

Addiction is extremely difficult to overcome. Not every overdose is because of treatment facilities, but the number of overdoses occurring is higher than it should be. A big issue people have found with treatment facilities is a one size fits all policy. Treatment is generalized to fit the masses without working with patients to find the root cause of their addiction. Without personalized care and attention, patients often drop out, or get discharged without gaining sufficient benefit. By monitoring engagement, treatment facilities will be able to give patients the attention they need to reengage and increase retention rates. If treatment facilities work to be more effective in successfully treating patients, we could prevent the significant number of overdoses among people who have ever been in treatment. We are not aiming for zero, just a significant decrease.

Montgomery County offers Vivitrol to those in the custody of the county. Vivitrol is a nonaddictive, nonnarcotic opioid antagonist to help people overcome addiction and prevent relapse. I believe this to be effective because we saw a significant number of people who overdose were arrested for something regarding substances. When people are struggling with addiction, we see a significant number of them end up in the care of people who have the opportunity to help, including a hospital after overdosing the first time, treatment facilities, and even police custody. It is important that we take advantage of that opportunity to really help them, give them options, support, and resources so they leave with better chances of survival than they had before. I would like to see these changes implemented in Silver Spring, Wheaton, Gaithersburg, Rockville, and Germantown so that we do not miss an opportunity to save a life.

Reflection

I struggled with processing large datasets. The data dictionaries were so large and tedious to sift through and the data files would sometimes be too big for my computer to handle. In hindsight I wish I had found data that offered API or I worked with a program that was better suited for wrangling those datasets. I wish I had allotted more time to working through the data dictionaries because they ended up holding me up from being able to continue on to my EDA. I would have liked to have included the Substance Use and Mental Health Service Survey which

provided insight to the type of treatment that was available in each state and how many people were receiving specific types of treatment. But the data dictionary was so ill organized and dense, and there were too many NAs it became a lost cause.I found it incredibly rewarding bringing all of this together into one cohesive data story and being able to make meaningful findings that could translate to recommendations to better help people.

Thank you so much to my amazing professors who guided me through this experience, Prof. Alraee, Prof. Perine, and Prof. Saidi. Thank you to my classmates for the feedback and help. And thank you to the Montgomery County government and DataMontgomery team.

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